

KAUSHALRAJSINH PUWAR

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EDUCATION

International Institute of Information Technology Bangalore (IIIT-B)

Jul 2023 – Present

Integrated M.Tech in Computer Science and Engineering

CGPA: 2.89/4.00

EXPERIENCE

OpenG2P

May 2026 – Present

Machine Learning Intern

- Built an **LLM-assisted pipeline** converting natural-language queries into schema-constrained SQL with deterministic validation, retry logic, and multi-source routing.
- Evolved the static pipeline into a **live-loop agent** with sequential tool calling, mid-investigation clarification, and adaptive strategy based on intermediate results.
- Enabled **cross-source analysis** across 4 database systems with structured output contracts, logging, and confidence scoring, reducing manual triage effort.

PROJECTS

Interactive Portfolio Website | React, Tailwind CSS, Framer Motion, OpenGL, Vite 📄 [code] | 🌐 [live] Apr 2026

- Designed a **dark editorial design system** with a gold-on-near-black palette, dual type scale (Cormorant Garamond + Inter), and reusable UI primitives for sections, cards, buttons, and form fields across a **7-section** single-page layout.
- Crafted layered interaction design: a **WebGL star-field background** with **multi-layer parallax** and cursor repulsion, plus **3D photo tilt**, gold glare-sweep, scroll-triggered reveals, and animated project filtering using Framer Motion springs.
- Shipped production UI polish with an accessible contact form (**inline validation**, ARIA live regions, loading/success/error states) and **device-aware rendering** that disables GPU-heavy hover effects on touch devices while preserving the full experience on desktop.

AutoDocLM | Python, Ollama, OpenAI SDK, ChromaDB, NetworkX, MkDocs 📄 [code]

Mar 2026 – Apr 2026

- Developed an **LLM-powered code understanding system** that transforms large repositories into structured documentation, architecture summaries, and Mermaid dependency diagrams.
- Built a static analysis pipeline using **AST parsing**, **graph analytics**, and **PageRank centrality** to model cross-file dependencies and prioritize high-impact code components.
- Integrated **retrieval-augmented generation (RAG)** with local embeddings, ChromaDB, and external inference APIs to generate architecture-aware technical documentation at repository scale.

Adaptive Bridge | ROS 2, Python, Fast DDS, Cyclone DDS, Docker 📄 [code]

Jan 2026 – Mar 2026

- Built a **proxy-based middleware** that isolates critical and noncritical ROS 2 subscribers via independent DDS writers, eliminating slow-subscriber backpressure.
- Reduced critical subscriber tail latency **from 15 s to 1.55 ms at p95 (~9,700x)** and preserved publisher throughput at **30 Hz** under Gilbert-Elliot bursty loss.
- Validated across Fast DDS and Cyclone DDS using a reproducible Docker-based testbed; wrote an IEEE RA-L manuscript targeting mixed-criticality robotics networking.

Grid Hero | C++, Java, C 📄 [code]

Oct 2024 – Nov 2024

- Developed a CLI grid-based strategy game with player navigation and enemy pursuit.
- Implemented shortest-path enemy bot and **multi-level progression mechanics**.
- Designed **3 progressive levels** and a final boss stage with dynamic enemy pathfinding.

TECHNICAL SKILLS

- Programming Languages:** Python, C/C++, Go, Rust, JS/TS, Java, HTML/CSS, PostgreSQL
- Libraries & Frameworks:** React, Spring Boot, Node.js, Tailwind, Matplotlib, PyTorch, NumPy, Hugging Face, Gradio, ROS 2
- Tools & Platforms:** MongoDB, Docker, AWS, Kubernetes, Jenkins, Anaconda, Bedrock, Jupyter Notebook, Android Studio, Linux

RELEVANT COURSEWORK

- | | | |
|--------------------------------|-----------------------|------------------------|
| • Data Structures & Algorithms | • Operating Systems | • Machine Learning |
| • Computer Networks | • Database Management | • Software Engineering |

EXTRACURRICULARS

AWS Student Builders Group | Core Member

Jun 2026 – Present

- Part of the core organizing team responsible for supporting AWS-focused workshops, technical events, and student community initiatives.

Infin8 Sponsorship Team | Sponsorship Coordinator

Nov 2023 – Feb 2024

- Contributed as the Sponsorship Coordinator for the annual cultural festival Infin8 and secured sponsorships for the event